RADE: The Recursive Axiom-Derivation Engine — Deduction over Prediction in a Persistent Truth Graph

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> *"Novelty by guessing and novelty by formal expansion look alike from the outside. Only one of them can show its work."*

Abstract

A Transformer-style attention mechanism remains a statistical prediction machine regardless of how its routing is rearranged: what is computed is a distribution over continuations conditioned on co-occurrence, and changing the plumbing does not change that. This paper specifies **RADE**, an architecture organised around **logical synthesis** rather than probabilistic approximation.

RADE replaces three components at once. Knowledge moves from floating-point weights into an **Axiom Substrate** — a persistent graph of verified relations. Inference moves from a forward pass to a **proof pass** over that graph. And termination moves from a fixed depth to **homeostatic inhibition**, a resource budget whose exhaustion triggers a lateral switch to a different region of the graph rather than an unbounded search.

The consequence that matters is a change of failure mode. A predictive system fails by

producing fluent falsehood, which is undetectable from the output alone. RADE fails by **logical inconsistency**, which is detectable, localisable, and attributable to specific premises.

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1. The Problem with Rearranged Plumbing

Architectural variation within the predictive family — sparse attention, linear attention, state-space substitutions, mixture routing — changes the cost of computing $P(w_n \mid w_1 \dots w_{n-1})$ without changing what that quantity is. A count of how often one symbol followed another is not an account of *why* it did, and no rearrangement of the machinery that computes the count converts it into one.

This is not an argument that prediction is useless. It is an argument about what a system can be asked for. A predictive system can be asked for a continuation. It cannot be asked for a *derivation*, because it does not have one to give — the output was not produced by steps that could be exhibited.

RADE is specified to answer the second kind of request.

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2. The Axiom Substrate

2.1 Knowledge as structure, not as parameters

In a Transformer, knowledge is distributed across floating-point weights and cannot be inspected, edited, or deleted at the granularity of a fact. In RADE it is held explicitly:

- **Formal objects** O — concepts, as graph vertices.
- **Operators** Φ — typed relations, as directed edges.

The distinction from embeddings is persistence semantics. An embedding degrades under continued training: the representation of a concept drifts as unrelated updates accumulate. A graph edge does not. If the substrate holds `A is_a B`, it holds it until something explicitly removes it, and the removal is an event that can be logged.

2.2 Why this matters for correction

A predictive model cannot be told that one specific thing it believes is wrong. The belief is not localised, so correction is retraining, and retraining perturbs everything else.

In a truth graph, a wrong fact is an edge. Removing it is $O(1)$, and the consequences of removing it are computable: every derived edge citing it as a premise becomes unsupported and can be retracted by following the provenance chain. Correction is therefore *bounded* rather than global.

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3. The Proof Pass

The forward pass is replaced by a traversal.

Input. A query — for example, *why is X a superconductor?*

Operation. The engine does not estimate an answer. It searches the graph for a path connecting the queried concepts through typed operators. If a path exists, the path *is* the answer, and every edge along it is either an asserted premise or a derived edge that names the two premises that produced it.

When no path exists. The engine enters a **search-and-constrain** phase: it proposes a candidate link and tests it against the existing axioms. A candidate that contradicts a held axiom is rejected. A candidate that survives is committed as a derived edge, with its provenance recorded.

This is the distinction the paper turns on:

> **Novelty by guessing** produces a plausible string. **Novelty by formal expansion** produces an edge that survived a consistency test and can name what it rests on.

Both look like new output. Only the second can be audited, and only the second can be retracted cleanly when its support is withdrawn.

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4. Homeostatic Inhibition

An unbounded proof search does not terminate on a large graph. The biological analogue of the fix is inhibitory interneurons, which prevent excitation from propagating without limit.

Resource budget. Every search path is allocated a bounded budget.

Lateral inhibition. A path that yields no result within its budget is abandoned, and the search is displaced to a different region of the graph rather than deepened.

The behavioural effect is the ability to *stop pursuing an unproductive line*, which is a capability rather than a limitation: a system that cannot abandon a path cannot allocate effort. This mechanism is the direct ancestor of the tension-bounded construction budget in the successor architecture, where the budget is derived from a physical observable rather than set as a constant.

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5. Comparison

Property	Transformer	RADE
Foundation	statistical probabilities	logical axioms
Inference	forward pass	proof pass over a graph
Novelty	interpolation, indistinguishable from hallucination	formal expansion, consistency-tested
Memory	lossy KV cache	persistent graph of hard relations
Correction	retraining, global	edge removal, bounded by provenance
Failure mode	fluent falsehood, **undetectable from output**	logical inconsistency, **detectable and localisable**
Objective	minimise prediction error	maximise logical consistency

The failure-mode row carries the argument. A system whose errors are detectable can be improved by an operator who does not have access to its internals. A system whose errors are fluent cannot.

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6. Ingestion: The Axiom-ifier

A corpus is not knowledge until it is structured. RADE separates **reading** from **reasoning**: an ingestion stage converts source material into structured triples, and the reasoning engine never touches raw text.

The ingestion stage emits only triples that the substrate can type-check against its operator set. Anything it cannot structure is dropped rather than approximated — a coverage gap is preferable to a fabricated relation, because a gap is visible and a fabrication is not.

The successor architecture implements exactly this split: mechanical AST extraction produces triples from source, and every fact carries a provenance class distinguishing what was asserted from what was derived and what was established by experiment.

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7. Status and Relation to the Implemented System

RADE is an **architectural proposal**, written before the substrate that now exists. Its components map onto implemented mechanisms as follows:

RADE component	Implemented as	Status
Axiom Substrate	typed concept graph, `data/*.axioms`	**live**
Proof pass	forward chaining with premise citation	**live** — derived edges name their parents
Search-and-constrain	contradiction check before assertion	**live**
Resource budget	tension-bounded construction budget	**live**, now derived from field energy
Lateral inhibition	budget exhaustion ends a derivation chain	**partial** — no region-switching heuristic
Axiom-ifier	mechanical AST extraction	**live**, no ML

The one component not carried across is the explicit region-switching behaviour of §4. The current engine exhausts a budget and stops; it does not redirect to a different subgraph. Whether that redirection is worth implementing is open, and it should not be reported as present.

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8. Limitations

Ingestion is the bottleneck, not reasoning. The proof pass is cheap; converting unstructured source into typed triples is not. The quality ceiling of the whole system is the quality of that conversion, and this paper does not solve it.

Consistency is not truth. The search-and-constrain phase rejects candidates that contradict held axioms. It cannot reject a candidate that is consistent with a set of axioms that are themselves wrong. Consistency is a necessary condition and nothing more.

Coverage cliff. A query whose concepts are absent from the substrate returns nothing. This is the correct behaviour and it is also a visible limitation; the honest presentation is an explicit gap rather than a degraded guess.

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9. Conclusion

RADE's claim is narrow and, stated carefully, defensible: an architecture that stores relations rather than parameters, infers by traversal rather than by prediction, and bounds its search by an explicit budget can answer for its outputs in a way a predictive system structurally cannot.

It does not claim to reason better. It claims to reason *auditably* — and to fail in a way that can be seen.

References

- Mahir, I. *From Artificial to Synthetic Intelligence: Principles of Constructive Structural Cognition.* — the tri-layer architecture in which RADE's components are implemented.
- Mahir, I. *JAS: Einstein's Epistemology as a Running Architecture.* — the provenance classes that make retraction bounded.
- Implementation: `src/rade.hpp`.